nf-core Fundamentals

From Nextflow concepts to reproducible, community-maintained bioinformatics pipelines.

What changes after Workshop 1?

Nextflow engine > **nf-core pipelines** > **Profiles HPC settings** > **Slurm tasks** > **Apptainer software**

Nextflow runs processes and channels; nf-core supplies ready-made, versioned pipelines. Institutional profiles connect those pipelines to Purdue Slurm and Apptainer.

Two types of profiles: do not confuse them

-profile apptainer enables container execution only (when the pipeline provides this profile). It does not configure Slurm accounts, partitions or resource routing.
-profile purdue_... loads an institutional profile that sets Slurm execution, Apptainer and HPC resource policies. On Purdue, prefer the matching institutional profile.

```
# Container-only profile: NOT a complete HPC configuration
nextflow run nf-core/demo -profile test,apptainer

# Institutional profile: Slurm + Apptainer + site policies
nextflow run nf-core/demo -profile test,PURDUE_PROFILE \
  --cluster_account YOUR_ACCOUNT --outdir results
```

Use test only for example datasets. Purdue institutional profiles include purdue_bell, purdue_gautschi and purdue_negishi; choose the one matching your system.

Scratch first

Start Nextflow from a compute allocation, not a login node. Keep work directories and container caches in $RCAC_SCRATCH; move important results to persistent storage.

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Control where Nextflow writes

Purdue uses Apptainer. Some Nextflow and nf-core settings still contain the historic word Singularity; this does not mean you must install Singularity.

```
mkdir -p "$RCAC_SCRATCH/nfcore-workshop"
cd "$RCAC_SCRATCH/nfcore-workshop"
mkdir -p work images apptainer-tmp apptainer-cache

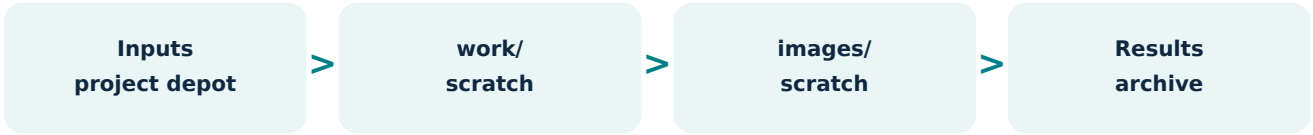
export NXF_HOME="$PWD/.nxf-home"
export NXF_WORK="$PWD/work"
export NXF_SINGULARITY_CACHEDIR="$PWD/images"
export NXF_APPTAINER_CACHEDIR="$PWD/images"
export APPTAINER_CACHEDIR="$PWD/apptainer-cache"
export APPTAINER_TMPDIR="$PWD/apptainer-tmp"
```

What does each variable mean?

NXF_HOME — Nextflow installation, plugins and metadata; avoid filling HOME quota.
NXF_WORK — hashed task directories, intermediate files and cache needed for -resume.
NXF_SINGULARITY_CACHEDIR — Nextflow-managed container image cache when the singularity engine is selected. Common in older Purdue examples.
NXF_APPTAINER_CACHEDIR — corresponding Nextflow cache for the native apptainer engine; use for Purdue institutional profiles that enable apptainer.
APPTAINER_CACHEDIR — Apptainer's own download/blob cache (different from Nextflow image cache).
APPTAINER_TMPDIR — temporary space for pulling and converting images.

> Important: An institutional profile can define apptainer.cacheDir itself. Check its documented configuration; explicit config settings can take precedence over environment defaults. Both cache names are shown for teaching, but you only need the one matching the engine in use.

Where data should live



Scratch is not backed up. Keep the input sheet, run command, config and final reports with archived results.

Customize a site profile safely

Institutional profiles provide sensible defaults, not fixed limits for every project. Use a small overlay config rather than editing the shared nf-core profile.

Example: custom.config

```
// Example overrides for a modest workshop run
process {
  withLabel: 'process_low' {
    cpus = 2
    memory = '4 GB'
    time = '2h'
  }
  withName: '.*:FASTQC' {
    cpus = 2
    memory = '4 GB'
  }
}
// Keep the institutional Slurm executor and Apptainer settings.
```

Labels and process names vary by pipeline. Inspect the pipeline source/config and use the exact names it defines. withName matches a process name; withLabel matches a label. Specific selectors can override more general settings.

Layer the override with -c

```
nextflow run nf-core/demo \
  -profile test,PURDUE_PROFILE \
  -c custom.config \
  --cluster_account YOUR_ACCOUNT \
  --outdir "$PWD/results" -resume
```

-c adds your custom configuration on top of the pipeline and institutional settings. Review the effective configuration before a large run:

```
nextflow config nf-core/demo -profile PURDUE_PROFILE \
  -c custom.config -flat | less
```

Choose CPUs, memory and time

Start with pipeline defaults. Check a representative task's trace.txt, Slurm seff JOBID, and memory/time failures. Increase only the processes that need it. A request that exceeds a node or partition limit can remain pending or fail submission.

> Resource principle: The manager job (e.g., 2 CPUs / 8 GB) is separate from pipeline tasks. The institutional profile determines how each task is submitted. Do not confuse manager #SBATCH settings with process.cpus / memory / time.

A repeatable HPC launch

```
#!/bin/bash
#SBATCH --job-name=nfcore-demo
#SBATCH --account=YOUR_ACCOUNT
#SBATCH --partition=YOUR_PARTITION
#SBATCH --cpus-per-task=2
#SBATCH --mem=8G
#SBATCH --time=04:00:00
#SBATCH --output=nfcore-%j.out

module load nextflow
cd "$RCAC_SCRATCH/nfcore-workshop"
export NXF_HOME="$PWD/.nxf-home"
export NXF_WORK="$PWD/work"
export NXF_APPTAINER_CACHEDIR="$PWD/images"
export APPTAINER_CACHEDIR="$PWD/apptainer-cache"
export APPTAINER_TMPDIR="$PWD/apptainer-tmp"
mkdir -p "$NXF_APPTAINER_CACHEDIR" \
  "$APPTAINER_CACHEDIR" "$APPTAINER_TMPDIR"

nextflow run nf-core/demo -profile test,PURDUE_PROFILE \
  -c custom.config --cluster_account YOUR_ACCOUNT \
  --outdir "$PWD/results" -resume -ansi-log false
```

Replace account, partition and profile; save as run_demo.sh, then run sbatch run_demo.sh. For a real pipeline, pin its release with -r VERSION and supply its required inputs.

Inspect, resume, and preserve

```
squeue -u "$USER"      # monitor jobs
nextflow log            # run history
# failed task: inspect work/xx/hash/.command.err
# rerun the same command with -resume
nextflow clean -n       # preview only; do not rush cleanup
```

Need help? nf-core Slack

Join nf-co.re/join. Ask in #help, #nostupidquestions or the pipeline-specific channel. Include the pipeline version, command, profile, relevant logs and sanitized samplesheet details. Contact Purdue RCAC for allocation, quota and scheduler problems.

> Take-home: Institutional profile = HPC integration. Custom -c = targeted overrides. Apptainer cache + Nextflow work = scratch. Final results = persistent storage.